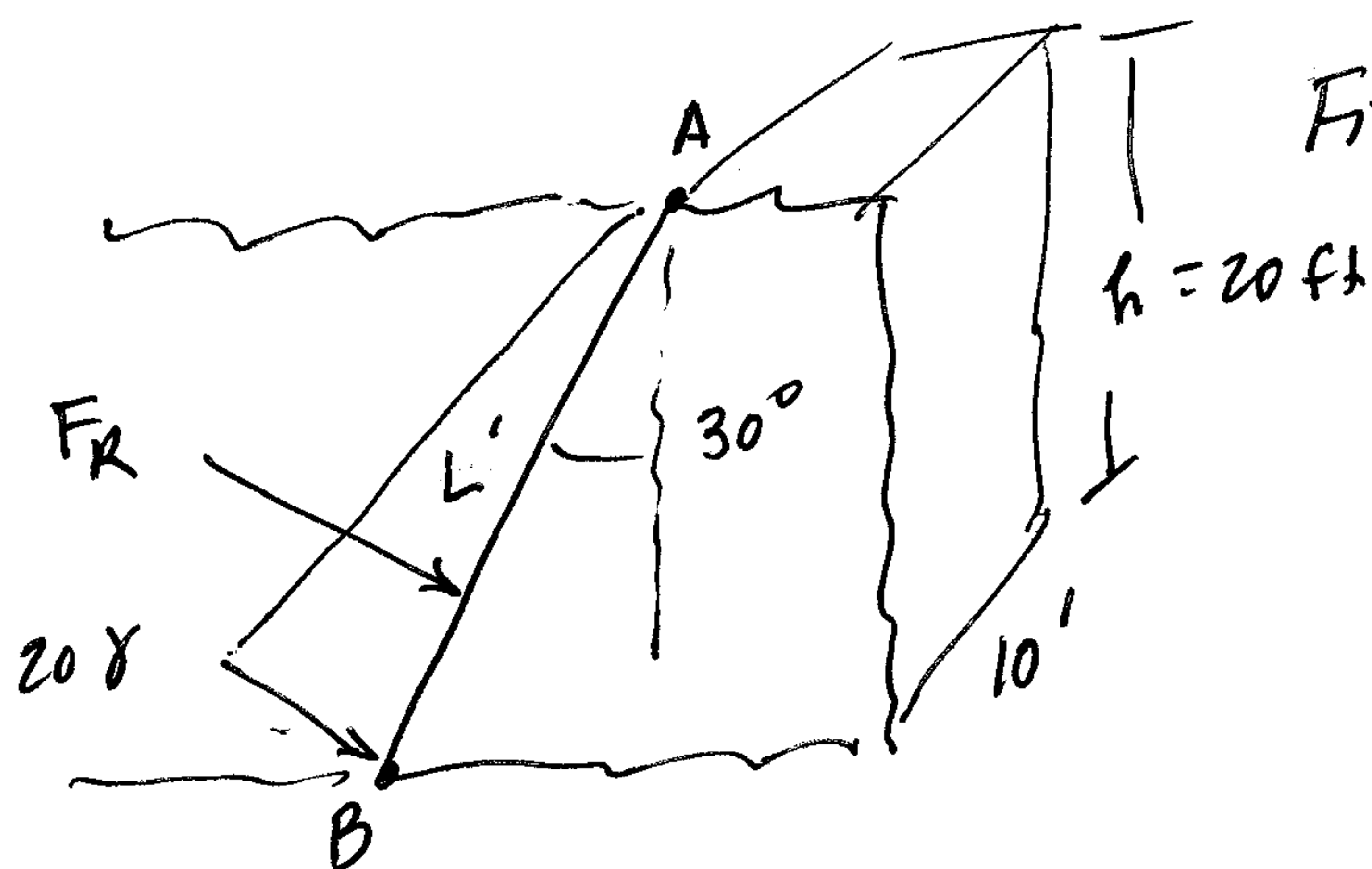


Simple Problem worked three ways



Find F_R .

Fresh Water:

$$\gamma = 62.4 \frac{\text{lb}}{\text{ft}^3}$$

$$L \cos 30 = 20$$

$$L = \frac{20}{\cos 30} = 23.09 \text{ ft}$$

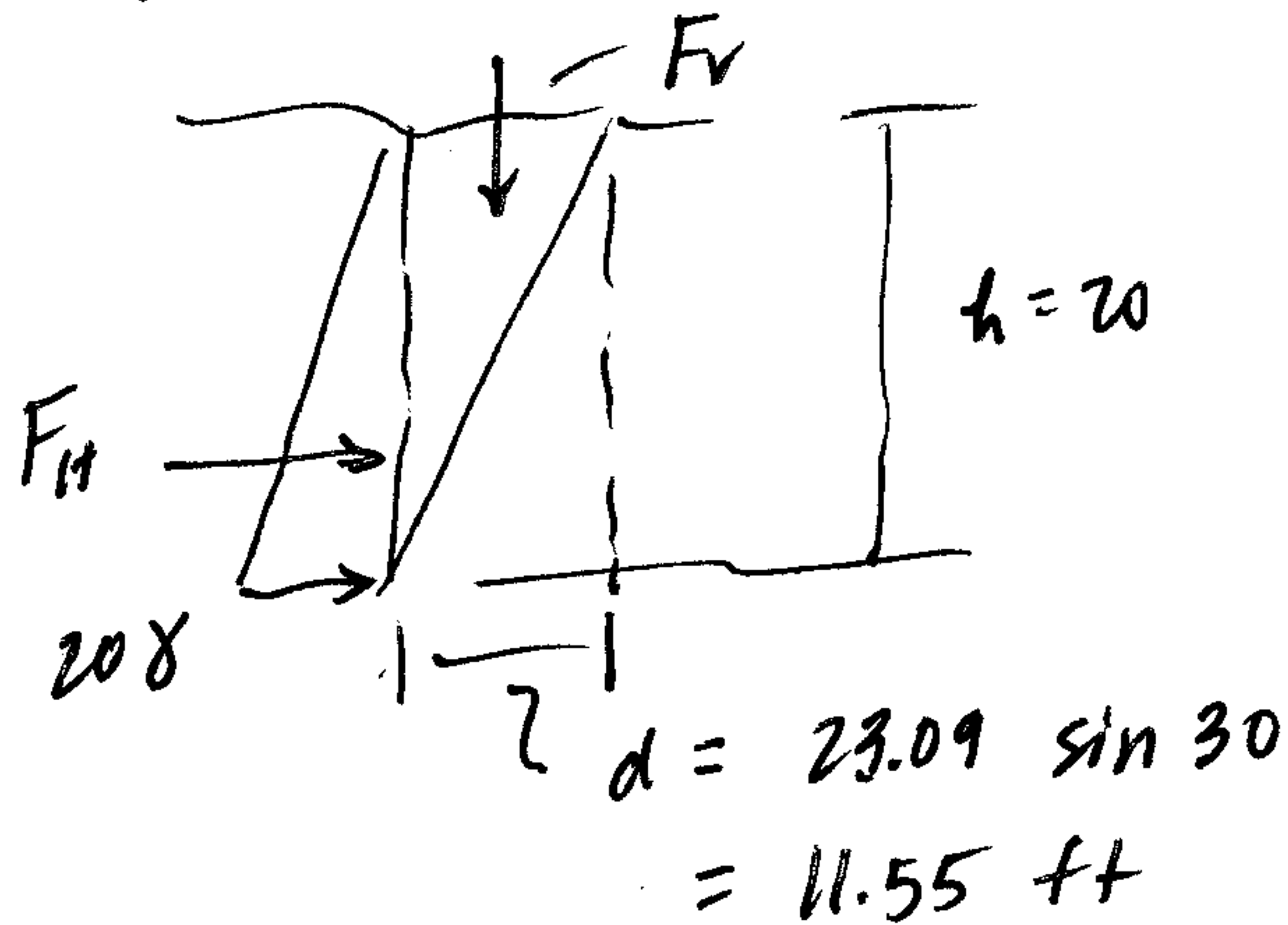
(a) Δ distribution:

$$F_R = \frac{1}{2} \underbrace{(208)}_b \underbrace{(23.09)}_h (10)'$$

$$= 144.1 \text{ kip at } \begin{array}{c} 30^\circ \\ \swarrow \end{array}$$

$$\text{Acts } \frac{1}{3}(23.09) = 7.70 \text{ ft up from B}$$

(b) $\vec{F}_R = \vec{F}_H + \vec{F}_V$



$$F_H = \frac{1}{2} (208) (20) (10') \quad \uparrow \text{ width!}$$

$$= 124.8 \text{ kip} \rightarrow$$

$$F_V = W = \frac{1}{2} (11.55) (20) (10) \gamma$$

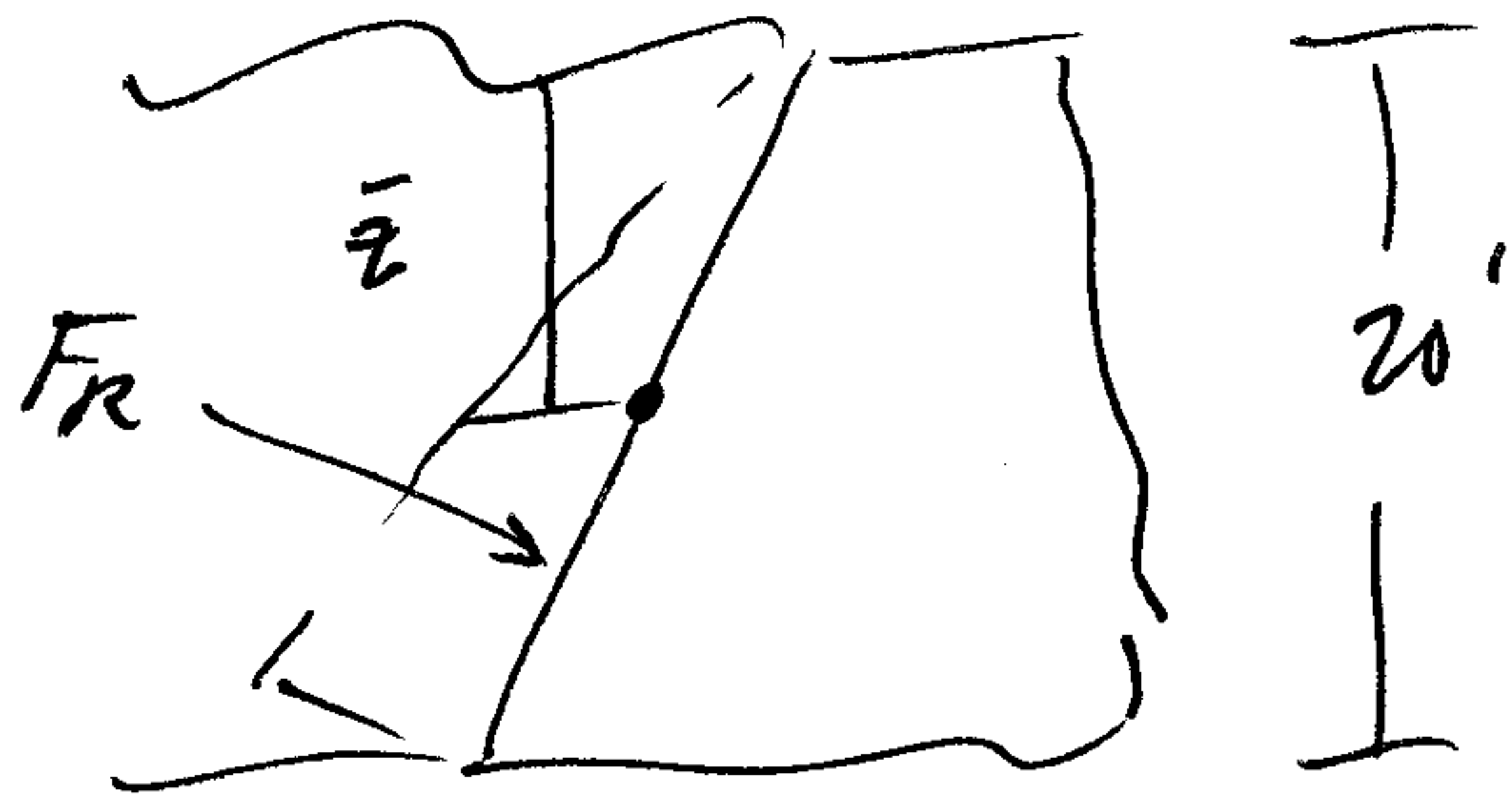
$$= 72.1 \text{ kip} \downarrow$$

$$\vec{F}_R = \sqrt{124.8^2 + 72.1^2}$$

$$= 144.1 \text{ kip} \quad \searrow 30^\circ$$

$$= [124.8 \hat{i} - 72.1 \hat{j}] \text{ kip}$$

(c) $F_R = \gamma \cdot \bar{z} \cdot A$ formula



$$\bar{z} = 10'$$

$$A = (23.09)(10)$$

$$F_R = \gamma \cdot \bar{z} \cdot A$$

$$= (62.4)(10)(23.09)(10)$$

$$F_R = 144.1 \text{ kip}$$